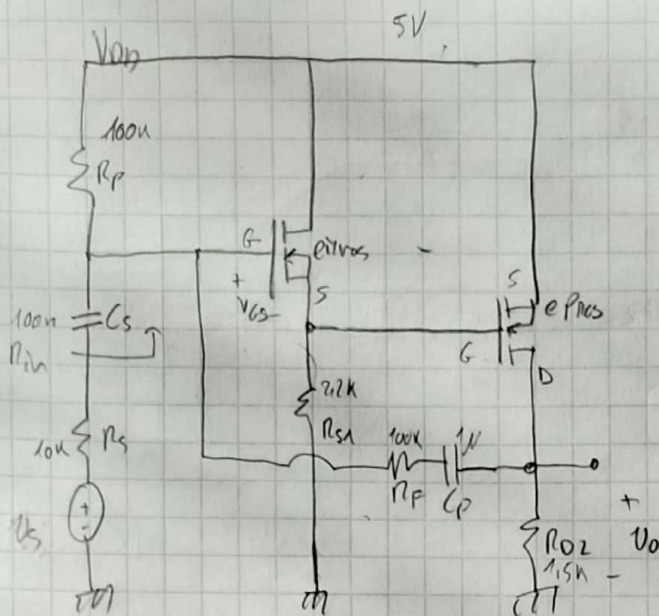




1) op point (V_{GS}, I_D)

$$\begin{cases} V_{DD} - V_{GS1} - R_{S1} I_{D1} = 0 \\ I_{D1} = I_{DSS} \left(1 - \frac{V_{GS1}}{V_T}\right)^2 \end{cases} \Rightarrow \begin{cases} I_{D1} = \frac{V_{DD} - V_{GS}}{R_{S1}} \\ I_{D1} = I_{DSS} \left(1 - \frac{V_{GS}}{V_T}\right)^2 \end{cases}$$

$V_{GS} = 1,33V$

$I_D = 1,653mA, V_{DS} = 1,33V$

$$V_{GS2} = -V_{GS1}$$

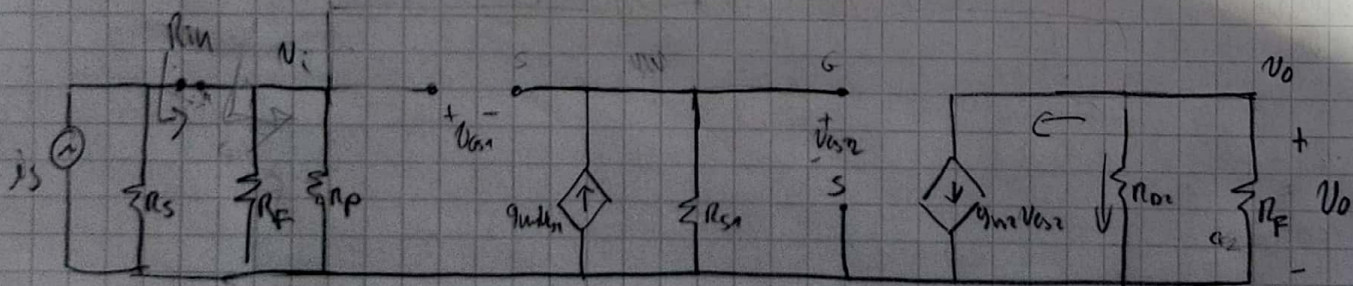
$$I_{D2} = I_{DSS2} \left(1 - \frac{V_{GS2}}{V_{T2}}\right)^2 = 1,653mA \quad V_{DS2} = R_{D2} I_{D2} - V_{DD} = -2,52V$$

$$g_{m1} = g_{m2} = 3,98mA/V$$

2) $P = V_{DD} (I_{D1} + I_{D2}) = 16,53mW$

3) $A_v = V_o / V_{is}$

Control voltage sampling $\beta = \frac{1}{R_F} \quad R_{in} = R_{out} = R_F = 15k\Omega \quad \frac{1/\beta}{1/\beta} = \frac{1}{R_S}$



$$A_v = \frac{V_o}{V_{is}} = \frac{V_o}{V_{GS2}} \cdot \frac{V_{GS2}}{V_i} \cdot \frac{V_i}{V_{is}}$$

$$\frac{V_o}{V_{os2}} = -g_m (R_{D2} || R_P) = -5,86$$

$$\frac{V_{os2}}{V_i} = \frac{g_m V_{os1} R_{S1}}{V_{os1} + g_m V_{os1} R_{S1}} = \frac{g_m R_{S1}}{1 + g_m R_{S1}} = 0,896$$

$$\frac{V_i}{i_s} = R_S || R_P || R_P = 8,57 \text{ k}\Omega$$

$$A_v^o = -44,997 \text{ k}\Omega \quad A_F = \frac{A_{vo}}{1 + \beta A_{vo}} = -34,61 \text{ k}\Omega$$

$$A_v = \frac{V_o}{V_s} = A_v^F \cdot \frac{1}{R_S} = \boxed{-3,46}$$

$$3) R_{in}^o = \frac{V_i}{i_s} = 8,57 \text{ k}\Omega \quad R_{in}^F = \frac{R_{in}^o}{1 + \beta A_{vo}} = \boxed{6,59 \text{ k}\Omega}$$

$$4) R_{out}^o = R_{D2} || R_P \quad R_{out}^F = \frac{R_{out}^o}{1 + \beta A_{vo}} = \boxed{11,14 \text{ k}\Omega}$$

7+L you can't apply feedback

→ from original circuit: - C_F sees at least $R_F + R_P || R_S \approx 160 \text{ k}\Omega$
- C_S sees $R_S + R_{in}$ calculated before

$$f_L = \frac{1}{2\pi} \left(\frac{1}{C_S R_{S1}} + \frac{1}{R_F \times C_F} \right) = 93 \text{ Hz}$$

$$\frac{V_{out}}{V_{in}}$$

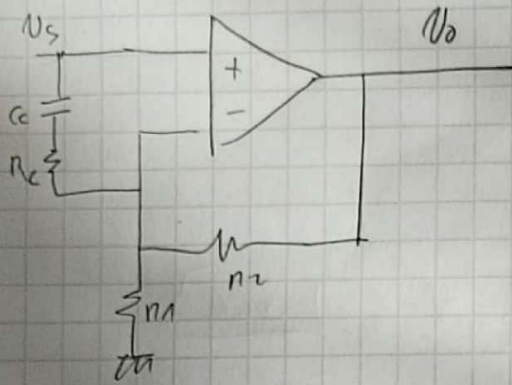
$$V_{out} = \frac{V_{in}}{1 + \beta A_{vo}} = \frac{V_{in}}{1 + 100 \times 100} = 93 \text{ Hz}$$

$$f_L = \frac{1}{2\pi} \left(\frac{1}{C_S R_{S1}} + \frac{1}{R_F \times C_F} \right) = 93 \text{ Hz}$$

$$f_L = \frac{1}{2\pi} \left(\frac{1}{C_S R_{S1}} + \frac{1}{R_F \times C_F} \right) = 93 \text{ Hz}$$

See also the other two cases (not shown)

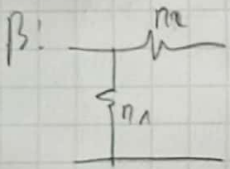
B2)



1) $A_v(s) = 1 + \frac{R_2}{R_1}$

2) feedback theory diagram.

Voltage source Voltage inputs

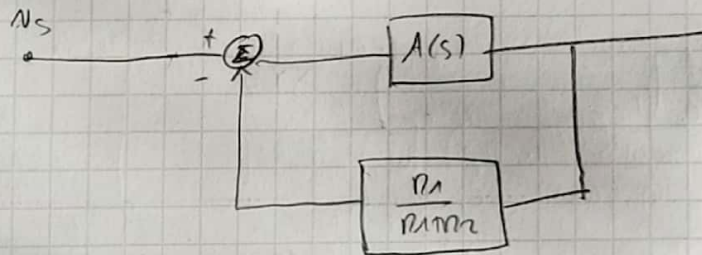


$R_{if} = R_1 || R_2$

$R_{of} = R_1 + R_2$

$\beta = \frac{R_1}{R_1 + R_2}$

$A_o^L = A(s) = \frac{U_o}{U_s}$



3) BW and PM

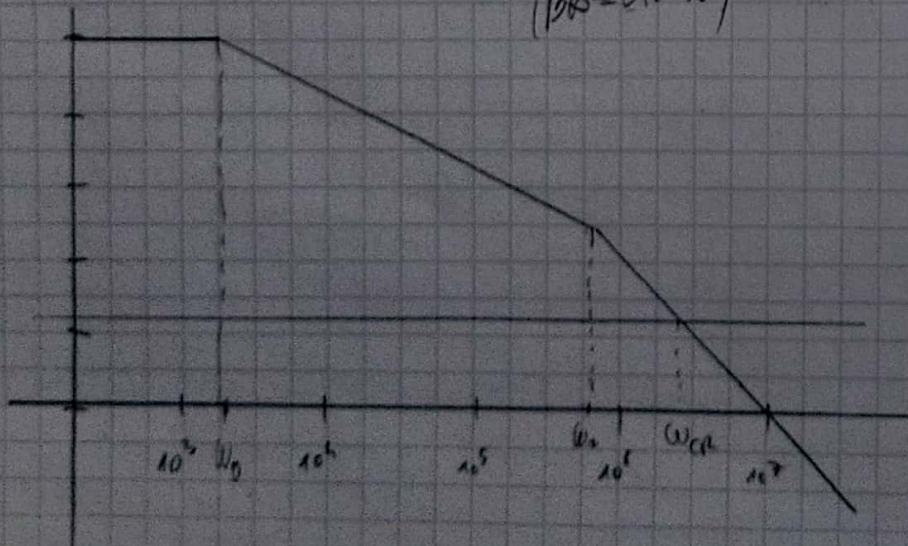
$B(s) = \frac{R_1}{R_1 + R_2}$

$\frac{1}{\beta_o} = \frac{1}{\beta_{po}} = 1 + \frac{R_2}{R_1} = 13,3 \text{ dB}$

$\omega_1 = 2 \cdot 10^3 \text{ rad/s}$

$\omega_2 = 6 \cdot 10^5 \text{ rad/s}$

$(\beta_{po} = 0,2136)$



$$A_0 \omega_0 = A_1 \omega_1$$

$$A_1 \omega_1^2 = 1/p_{10} \omega_{cn}^2$$

$$\omega_{cn} = \sqrt{A_0 \omega_0 \omega_1 p_{10}} = 5,086 \cdot 10^6 \text{ rad/s}$$

$$\varphi_m =$$

$$\varphi_m = 180 - \arctan\left(\frac{\omega_{cn}}{\omega_0}\right) - \arctan\left(\frac{\omega_{cn}}{\omega_1}\right) = \boxed{6,75^\circ}$$

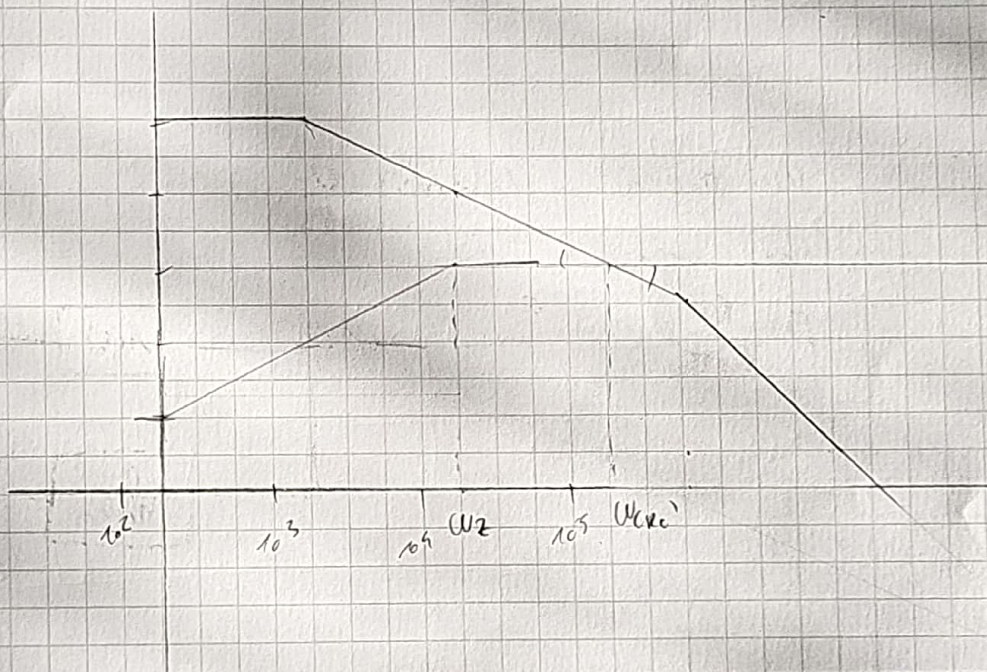
4) with CC, RC, new BW and pm

$$B(s) = \frac{p_{11} \frac{1+sC_cR_c}{sC_c}}{R_2 + p_{11} \frac{1+sC_cR_c}{1+sC_c}} = \text{complex. we know the zero location } \omega_z = \frac{1}{C_c R_c} = 2 \cdot 10^4 \text{ rad/s}$$

$$B_0 = \frac{p_{11} R_c}{R_2 + p_{11} R_c} =$$

$$\frac{1}{B_0} = 58 \text{ dB}$$

$$\frac{1}{B_0} = 13,3 \text{ dB}$$



$$A_0 \omega_0 = 1/p_{10} \omega_{cn} \quad \omega_{cn} = p_{10} A_0 \omega_0 = 2,48 \cdot 10^5 \text{ rad/s}$$

ω_p is so low it counts for -90° in φ_m

$$\varphi_{nc} = 180 - 90 - \arctan\left(\frac{\omega_{cn}}{\omega_0}\right) - \arctan\left(\frac{\omega_{cn}}{\omega_1}\right) + \arctan\left(\frac{\omega_{cn}}{\omega_z}\right) = \boxed{63,4^\circ}$$

1h 5mn